Latest version: `https://vuphan314.github.io`
This version: 2019/12/25

Vu H. N. Phan

CV

vuphan314

Education

2017/08–present **Doctor of Philosophy**, *Computer Science*, Rice University, Houston TX.
- ***Advisor: Moshe Vardi***
- Research: weighted model counting (propositional logic)
- GPA: 3.60

2014/08–2017/08 **Dual Bachelor of Science**, *Computer Science & Math*, Texas Tech University, Lubbock TX.
- GPA: 4.00

2013/01–2014/08 **Associate of Science**, Collin College, Plano TX.
- GPA: 4.00

Research

Published

2018–2020 **Weighted Model Counting with Algebraic Decision Diagrams**, *Computational Logic*.
- Developed the model counter `ADDMC` using dynamic programming and early variable projection with Moshe Vardi and Jeffrey Dudek.
- Presenting our paper at the AAAI conference in 2020.

Supervisor: Moshe Vardi (Computer Science Department, Rice University).

2016–2018 **Syntactic Conditions for Antichain Property in `CR-Prolog`**, *Logic Programming*.
- Found syntactic conditions guaranteeing when `Consistency Restoring Prolog` programs have antichain property (no answer set is a proper subset of another).
- Presented my paper at the ASPOCP workshop (affiliated with ICLP, part of FLoC) at the University of Oxford on 2018/07/18.

Supervisor: Dr. Michael Gelfond (Computer Science Department, Texas Tech University).

Unpublished

2013–2017 **Biquasiarithmetic Mean Comparability**, *Mathematical Analysis*.
- Generalized a theorem on inequalities involving quasiarithmetic means.
- Resubmitted my paper to Rose-Hulman Undergraduate Math Journal in 2017 (previously submitted in 2013 & 2014).
- Presented the results at the 2013 Texas Undergraduate Math Conference.

Supervisor: Dr. Alan Graves (Math Department, Collin College).

Work

Software Developing

2016–2017 **LED**, *Literate Programming System*.

- Implemented the Language of Effective Definitions by automating its translation into the functional programming language SequenceL.
- Implemented an LED-to-TeX translator.

Supervisor: Dr. Nelson Rushton (Computer Science Department, Texas Tech University).

2016 **Knotty**, *Computer Algebra System*.

- Designed the procedural programming language Knotty for knot theory.
- Implemented Knotty by automating its symbolic computation and TeX printing.
- Implementing additional knot-theoretic operations in Knotty.

Supervisor: Dr. Razvan Gelca (Math Department, Texas Tech University).

2015–2016 **L**, *Logic Programming Engine*.

- Implemented the language L by automating its translation into Answer Set Prolog.

Supervisor: Dr. Nelson Rushton (Computer Science Department, Texas Tech University).

Tutoring

2015, 2017 **Computer Science, Math, Physics, Chemistry**.
Supervisor: Mr. Patrick Bohn (Learning Center, Texas Tech University)

Scholarships

2017–present **Ph.D. Full Tuition Waiver & Stipend**, *Computer Science Department*, Rice University.
2016–2017 **General Scholarship**, *Financial Aid Office*, Texas Tech University.
2015–2017 **Academic Achievement Scholarship**, *College of Arts & Sciences*, Texas Tech University.
2015–2017 **Undergraduate Achievement Scholarship**, *Math Department*, Texas Tech University.
2015–2016 **Undergraduate Research Scholarship**, *Math Department*, Texas Tech University.
2014–2016 **Proven Achiever Scholarship**, *Financial Aid Office*, Texas Tech University.
2013–2014 **Excellence Scholarship**, *Math Department*, Collin College.

Courses

2019 Fall **Computer Systems Architecture**.
2019 Spring **Compiler Construction; Multi-Core Computing**.
2018 Fall **Artificial Intelligence**.
2018 Spring **Principles of Programming Languages; Automated Program Verification**.
2017 Fall **Advanced Logic in Computer Science**.

Computer Skills

- Programming Languages:
  - Procedural: Knotty (my design & implementation), Python/Java/C++/C, Assembly
  - Logic: L (my implementation), SPARC/Clingo/DLV
  - Literate: LED (my implementation)
  - Functional: SequenceL, Scala, Scheme
  - Formatting: TeX, HTML
  - Data: SQL
- Tools: Git/GitHub, Linux/Windows shells